

# Gaming the Two Numbers They Control: Quantization, Weight-Cut Bunching, and the Limits of Strength in 3.9 Million Powerlifting Results

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**Abstract**—A powerlifting competitor controls exactly two numbers: the weight loaded on the bar and their bodyweight at weigh-in. We ask whether these choices leave measurable statistical patterns, using  $\sim 3.94$  million competition results from the public-domain OpenPowerlifting database. We report four findings, each answered with a course-appropriate test and led by effect size (at  $n \approx 3.9\text{M}$  almost everything is “significant”). (i) *Quantization*: 96.2% of 13.4M attempt loads sit exactly on the 2.5 kg plate grid; a generalized likelihood-ratio test (GLRT) whose null parameter lies on the boundary is calibrated by a parametric bootstrap. (ii) *Weight-cut bunching*: bodyweight piles up just below every IPF class limit (de-heaped  $\log(\text{below/above}) = +1.92$  at 83 kg); a Cattaneo–Jansson–Ma density-discontinuity test rejects at six of the seven limits (surviving Holm and Benjamini–Hochberg); the lone exception is the 120 kg boundary of the open-ended 120+ class, where the incentive to cut is weakest. A non-limit control shows no bunching, and placebos show no bunching-direction false positives. (iii) *Allometry*: strength scales with bodyweight differently by sex ( $b = 0.75$  men,  $0.51$  women vs. the isometric  $2/3$ ; a  $\text{Sex} \times \log(\text{BW})$  interaction is highly significant and persists on common support). (iv) *Prediction*: bodyweight, sex, equipment, and age predict the total ( $R^2 = 0.70$ , random forest, leakage-safe grouped CV). Together, the two numbers competitors control leave clear statistical fingerprints, discreteness on the bar and strategic mass below the limit, while the scaling and extreme-value analyses find no evidence of a population strength ceiling. Code and data: <https://github.com/adirelm/opl-stat-theory>.

## I. INTRODUCTION

Powerlifting comprises three lifts (squat, bench press, dead-lift), each with three attempts, contested within bodyweight classes. Two features make it useful for studying strategic behaviour. First, the implements are *discrete*: plates come in fixed denominations, so the load a competitor can place on the bar is not a continuous choice. Second, eligibility is governed by a *threshold*: a lifter who weighs even slightly above a class limit is bumped into a heavier, stronger field. We focus on the two numbers a competitor records: the load chosen for each attempt, and the bodyweight at weigh-in. Both interact with a sharp administrative rule. We ask whether these two choices leave detectable statistical signatures, and what  $\sim 3.94\text{M}$  results reveal about the limits of human strength.

**Related work.** Manipulation of a “running variable” around an administrative threshold produces *bunching*, a standard tool in public economics [1] that is classically tested with the McCrary density-discontinuity test [2] and its modern, bias-corrected local-polynomial form due to Cattaneo, Jansson, and Ma [3]. Rapid pre-weigh-in weight cutting is well documented

in weight-class sports [4]. On the physiological side, the square–cube law predicts that strength, governed by muscle cross-sectional area, scales with bodyweight as  $\propto \text{BW}^{2/3}$ . Extreme-value theory has been used to estimate the ultimate limits of athletic performance from record data [5]. On OpenPowerlifting specifically, recent work studies *scoring and scaling* systems [6]; to our knowledge, no formal manipulation test of weight-cut bunching has been applied to this dataset.

**Contribution.** We organize the paper around the two recorded choices, load and weigh-in bodyweight, and use both halves of the course, classical inference and machine learning. Our contributions are: (1) a rounded-likelihood *quantization* GLRT for the plate grid, with a parametric-bootstrap calibration that respects the boundary nature of the null; (2) to our knowledge the first formal density-discontinuity test of weight-cut *bunching* on OpenPowerlifting, checked using a non-limit control, placebo cutoffs, de-heaping, and multiple-testing corrections; (3) a sex-interaction *allometry* model that tests the scaling exponent against the isometric  $2/3$ ; and (4) a leakage-safe strength-*prediction* model. Throughout we lead with effect sizes and confidence intervals rather than  $p$ -values, as explained next.

**Methodological stance.** At  $n \approx 3.9\text{M}$  the power to detect any non-zero effect is effectively one, so a  $p$ -value answers a question we do not care about (“is the effect exactly zero?”) rather than the one we do (“is it large?”). We therefore report effect sizes with 95% confidence intervals as the primary evidence, use  $p$ -values only as a secondary screen, and apply multiple-testing corrections to every pre-declared family: all four (Bonferroni, Holm, Šidák, Benjamini–Hochberg) for the H2 limits, and Holm for H3. Because the same lifter appears in many rows, naive standard errors understate uncertainty (pseudo-replication); the formal tests therefore use one row per lifter (with heteroskedasticity-robust HC3 errors), the deduplication unit declared per hypothesis.

## II. RESULTS

All intervals are 95%. H1 uses attempt loads; the cross-sectional tests use one row per lifter (to respect independence), and where the class scheme matters the modern kg-only era (2014+). Headline effect sizes use all rows.

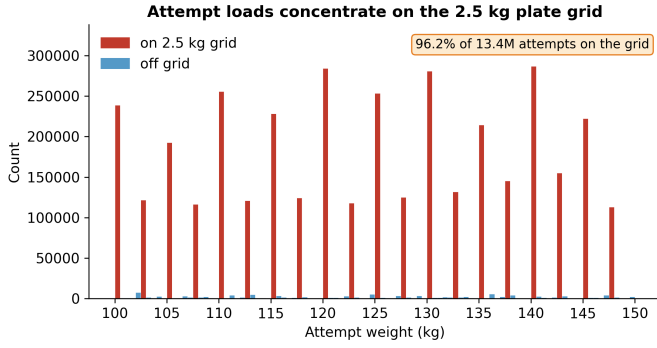


Fig. 1. H1: attempt loads concentrate on the 2.5 kg plate grid.

#### A. Attempt loads are quantized to the 2.5 kg grid (H1)

Of 13,397,702 valid attempt loads, **96.20%** (CI [96.19, 96.21]) fall exactly on the 2.5 kg plate grid (Fig. 1); the smallest standard plate is 1.25 kg, loaded on both sides, so 2.5 kg is the minimum increment. To test this formally we restrict to the kg-native (0.5 kg) lattice and model the within-cell position as a mixture that places weight  $\pi$  on the grid point and  $1 - \pi$  on a uniform background. The GLRT of  $\pi = 0$  versus  $\pi > 0$  has its null parameter on the boundary of  $[0, 1]$ , so Wilks' theorem fails: the null distribution of  $2 \ln \Lambda$  is the 50:50 mixture  $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$  of Chernoff and Self-Liang [7], [8]. Our parametric bootstrap closely matches this theory (point mass 0.51 vs. the theoretical 0.50; calibrated 95% cutoff 2.59 vs. the theoretical 2.71, both well below the naive  $\chi_1^2$  value of 3.84). The observed statistic,  $2 \ln \Lambda = 3.9 \times 10^7$ , exceeds every bootstrap draw ( $p < 3.4 \times 10^{-4}$ ), and a  $\chi^2$  goodness-of-fit against a uniform within-cell distribution strongly rejects it ( $5.0 \times 10^7$ ,  $\text{df} = 4$ ). The 3.8% off-grid loads are largely a units artifact: of the 2.1% that fall off the 0.5 kg lattice entirely, 82% are round pound loads (e.g., 102.06 kg = 225 lb), themselves quantized on a different grid.

#### B. Bodyweight bunches below class limits (H2)

Bodyweight piles up just below real class limits (Fig. 2). On de-heaped data (exact round/half-kilogram weigh-ins removed, so the signal is not mere digit preference), the log-ratio of just-below to just-above counts is positive at all seven IPF men's limits and strongest at 83 kg ( $+1.92 \pm 0.04$ , a  $6.8\times$  asymmetry; Table I, Fig. 3); a non-limit control at 91 kg is flat-to-slightly-negative ( $-0.21$ ). The formal Cattaneo–Jansson–Ma density-discontinuity test [3] (one random meet per lifter, de-heaped, 2014+) rejects at six of the seven limits with the density dropping at the cutoff ( $t < 0$ ), surviving both Holm and Benjamini–Hochberg correction; the lone exception is the 120 kg cutoff ( $t = -1.6$ ,  $p = 0.11$ ), adjacent to the unbounded 120+ class where the incentive to cut is weakest. The control shows no excess below (a small *opposite*-direction discontinuity), and no placebo shows the bunching pattern. The effect is robust across equipment (raw  $+2.14$ , equipped  $+1.51$ ) and concentrated in the tested majority ( $+1.93$ ; the smaller untested subset reverses to  $-0.30$ , consistent with weight-

TABLE I  
DE-HEAPED  $\log(\text{BELOW}/\text{ABOVE})$  AT EACH IPF MEN'S LIMIT (ALL ROWS; COUNTS OVER THE 0.5 KG WINDOWS JUST BELOW/ABOVE EACH LIMIT).  
LAST ROW IS A NON-LIMIT CONTROL.

| Limit (kg)   | log-ratio    | $\pm$ | $\times$ asymmetry |
|--------------|--------------|-------|--------------------|
| 59           | +0.99        | 0.04  | 2.7                |
| 66           | +1.18        | 0.03  | 3.2                |
| 74           | +1.00        | 0.03  | 2.7                |
| <b>83</b>    | <b>+1.92</b> | 0.04  | <b>6.8</b>         |
| 93           | +1.65        | 0.04  | 5.2                |
| 105          | +1.17        | 0.05  | 3.2                |
| 120          | +0.74        | 0.06  | 2.1                |
| 91 (control) | -0.21        | 0.03  | 0.8                |

cutting being driven by the tested IPF-scheme federations that dominate this population), and it persists on the clean kg-only subset ( $+1.80$ ).

**Robustness and falsification.** The result survives four checks. (i) *De-heaping*: removing exact round/half-kilogram weigh-ins leaves the asymmetry intact, so the signal is not digit preference and is consistent with weight-cut manipulation. (ii) *Control*: the non-limit point at 91 kg shows no excess below. (iii) *Placebos*: cutoffs placed between real limits show no bunching-direction discontinuity after correction. (iv) *Subgroups and correction*: the effect holds within raw, equipped, and tested subsets, and the formal density test rejects at six of the seven limits even under the strictest (Bonferroni) correction.

#### C. Allometric scaling differs by sex (H3)

Fitting  $\log(\text{Total}) = a + b \log(\text{BW})$  (Fig. 4), the exponent is  $b = 0.75$  (CI [0.74, 0.75]) for men and 0.51 ([0.50, 0.51]) for women, with heteroskedasticity-robust (HC3) standard errors on one row per lifter. A two-sided Wald test rejects the isometric  $b = 2/3$  for both sexes, and in *opposite* directions (men  $z = +46.8$ , i.e.  $b > 2/3$ ; women  $z = -64.1$ , i.e.  $b < 2/3$ ). A pooled  $\text{Sex} \times \log(\text{BW})$  interaction estimates the men-minus-women slope gap at  $+0.24$  ( $p \ll 10^{-3}$ ). The gap also *grows* on the common bodyweight support (0.84 men vs. 0.38 women), so it is not an artifact of the sexes occupying different weight ranges. For men, the Pearson correlation of  $\log \text{BW}$  and  $\log \text{Total}$  is 0.60 (Fisher-transform CI [0.600, 0.603]; Spearman 0.58); it is lower for women. The message is not that women are weaker, but that the *estimated scaling* differs; we leave the mechanism to future work.

#### D. Strength is predictable from basic variables (H4)

Predicting the total from controllable/basic variables (bodyweight, sex, equipment, age), on one random meet per lifter subsampled to a seeded 200,000 lifters for tractability, with every split grouped by lifter and all preprocessing fit inside each training fold, a random forest reaches  $R^2 = 0.70$  (RMSE 86 kg) versus 0.61 (RMSE 98 kg) for linear regression (Fig. 5); permutation importance is dominated by sex and bodyweight, and continuous-feature VIFs are  $\approx 1$  (no collinearity concern). A logistic “made-weight” classifier built from *non-bodyweight*

## Bodyweight bunches just below a real class limit, not at a non-limit control

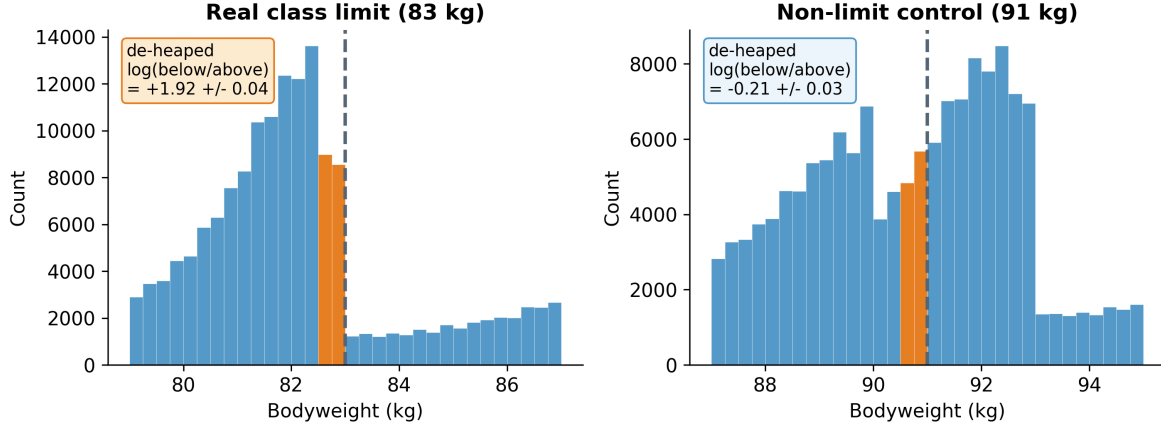


Fig. 2. H2: bunching just below a real limit (83 kg) but not at a non-limit control (91 kg); round-number weigh-ins removed.

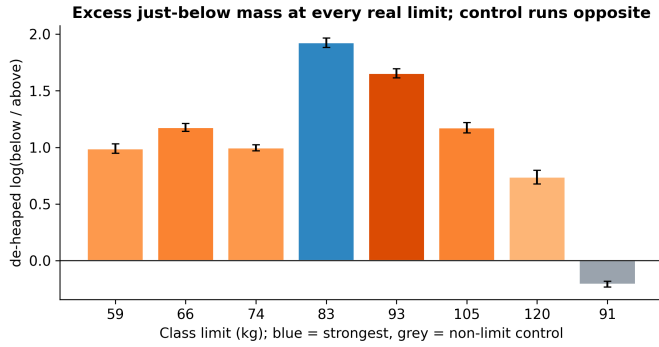


Fig. 3. H2: excess just-below mass at every real limit; the control runs opposite.

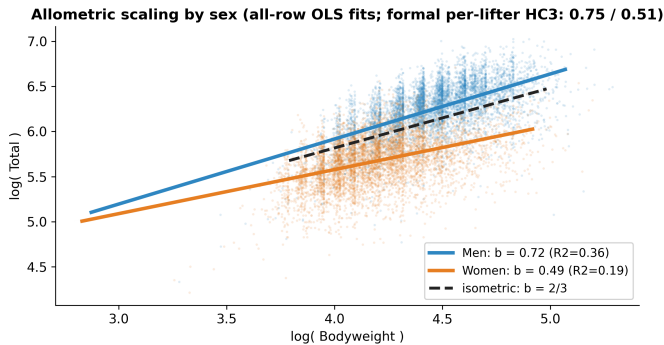


Fig. 4. H3: allometric scaling by sex vs. the isometric 2/3. Plotted lines are all-row OLS fits; the headline per-lifter HC3 exponents are 0.75 (men) / 0.51 (women).

features (age, tested status, era, equipment; the Dots score is excluded because it is a function of total and bodyweight) reaches AUC = 0.56 on  $n = 89,882$  lifters (PR-AUC 0.84 against a 0.82 no-skill baseline, i.e. the below-limit share; balanced accuracy 0.53). The signal is weak, as expected: cutting is driven by bodyweight relative to the limit, which

defines the label and which these features omit.

### E. Supporting analyses

**Distribution structure.** The apparent two-component shape of raw totals is just sex: by BIC and a bootstrap likelihood-ratio test, raw Total prefers two Gaussian components ( $\Delta\text{BIC} = 1.4 \times 10^3$ ) but the sex-normalized Dots score prefers one (Fig. 6); the two-component shape largely disappears after Dots normalization. **Tested vs. untested.** Untested competitions show higher Dots than tested ones (medians 344 vs. 318, Mann–Whitney  $p \rightarrow 0$ ; a parametric Welch test agrees,  $|t| = 128$ ; rank-biserial  $-0.20$ ), consistent with (though not proof of) an effect of performance-enhancing drugs, since tested and untested federations also differ in era, equipment, and selection. **Paired attempts.** Within a meet, the third squat attempt exceeds the opener by a median 15 kg (medians 160  $\rightarrow$  180;  $n = 10^5$  pairs; Wilcoxon signed-rank  $p \rightarrow 0$ ), the paired-sample counterpart of the two-sample comparison above. **Equipment.** Dots differs across equipment categories (ANOVA  $F = 5.3 \times 10^3$ ; Kruskal–Wallis agrees,  $H = 1.2 \times 10^4$ ; all six Bonferroni post-hoc pairs reject), with multi-ply highest (median Dots 401 vs. 328 raw). **Extreme values.** A GEV fit to equal-size block maxima of recent totals gives shape  $\xi = -0.02$  with bootstrap CI  $[-0.20, 3.77]$ : a population strength *ceiling is not supported*, since the CI includes zero. **Time trend.** Median Dots drifted down over 2000–2024 (Spearman  $\rho = -0.54$ ); one possible explanation is broadening participation, but the trend is not by itself evidence of declining strength. **Sequential detection and power.** A Wald sequential probability ratio test on the chronological stream of just-below/above indicators accepts the bunching hypothesis after a stopping time of only  $N = 27$  (Fig. 7). An a-priori calculation for the same design ( $p = 0.6$  vs. 0.5) requires  $n = 211$  for 0.90 power at  $\alpha = 0.05$ , and makes the Type I/II tradeoff concrete: at fixed  $n = 100$ , tightening  $\alpha$  from 0.10 to 0.01 raises  $\beta$  from 0.23 to 0.63. **Independence.** After deduplicating to one row per lifter, the

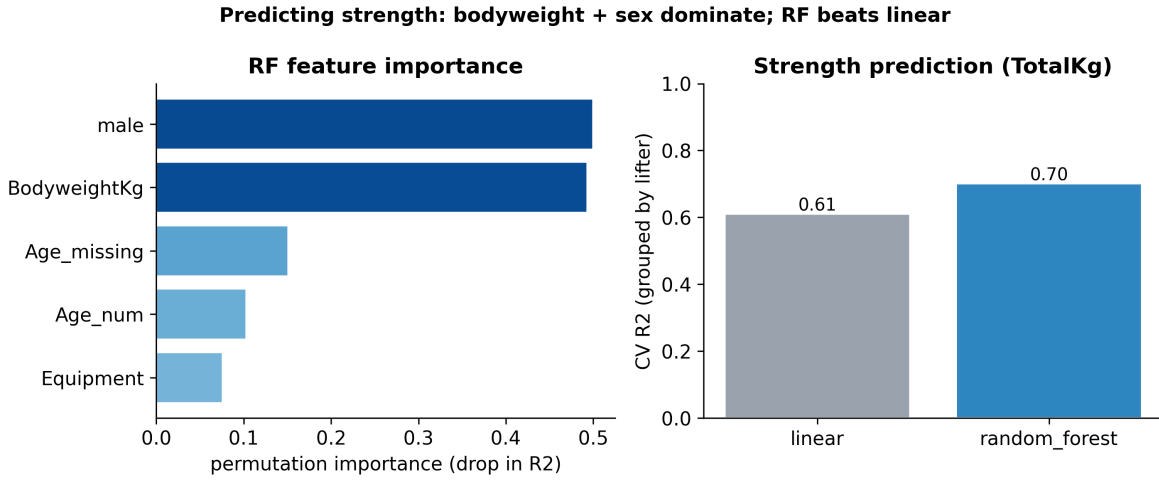


Fig. 5. H4: strength prediction; importance and grouped-CV  $R^2$ .

association between tested status and cutting is negligible (Cramér's  $V = 0.008$ ), illustrating how pseudo-replication inflates significance. **Residual normality.** The H3 (all-row fit) residuals are left-skewed with a heavy lower tail (skew  $-0.77$ ; Fig. 8), so they are not exactly normal. This does not undermine the slope: at this  $n$  its sampling distribution is asymptotically normal under standard conditions, and we report HC3-robust standard errors. A formal normality test rejects even in modest subsamples ( $n \approx 300$ ), as expected for genuinely non-normal residuals, so we treat it as a diagnostic and lead with effect sizes.

### III. METHODS

**Data and preprocessing.** We use a pinned snapshot of OpenPowerlifting [9] (3,941,811 rows, public domain; the SHA-256 is recorded for exact reproducibility). The unit of analysis is the *attempt* for H1 (the mechanism is within-competitor, so the grid is robust to clustering; attempts do cluster within lifters, so the attempt-level CI is optimistic in width and the conclusion rests on the effect size, 96% on-grid vs. the 20% no-preference baseline) and one *row per lifter* for the cross-sectional tests; *de-heaping* drops exact round/half-kilogram weigh-ins so the bunching signal is not digit preference. For prediction, all cross-validation is grouped by lifter so the same athlete never appears in both train and test.

**GLRT and the boundary (H1).** With log-likelihood  $\ell$ , the statistic is  $2 \ln \Lambda = 2(\ell_1 - \ell_0)$ . Under regularity  $2 \ln \Lambda \rightarrow \chi^2$ , but when the null fixes a parameter on the boundary of its space the limit becomes a mixture  $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$  [7], [8]; we estimate the null distribution directly by simulating data under the fitted null model (parametric bootstrap) and comparing the observed statistic to the simulated quantiles. Concretely, for within-cell position  $u \in \{0, \dots, 4\}$  on the 0.5 kg lattice the model is

$$P(u) = \pi \mathbf{1}[u = 0] + (1 - \pi) \frac{1}{5}, \quad \pi \in [0, 1], \quad (1)$$

and the GLRT contrasts  $\pi = 0$  (no grid preference, on the boundary) with the constrained MLE  $\hat{\pi} \geq 0$ .

**Density discontinuity (H2).** For a cutoff  $c$  the manipulation test contrasts the one-sided densities,  $\theta = \ln \hat{f}^+(c) - \ln \hat{f}^-(c)$ , estimated by local polynomials without pre-binning [2], [3];  $\theta < 0$  ( $t < 0$ ) indicates excess mass just below  $c$ . We complement the formal test with a de-heaped log count-ratio as the interpretable effect size, counting weigh-ins over the windows  $(c - 0.5, c]$  and  $(c, c + 0.5]$  kg.

**Allometry (H3).** OLS on log-log data gives the scaling exponent  $b$ ; we report HC3-robust standard errors (one row per lifter), a Wald statistic  $W = (\hat{b} - 2/3)/\widehat{\text{se}}(\hat{b})$ , and a pooled  $\text{Sex} \times \log(\text{BW})$  interaction whose coefficient is exactly the men-minus-women slope difference. Correlations use the Fisher  $z$ -transform for confidence intervals.

**Learning and the rest of the toolbox.** Prediction uses linear and random-forest regression and logistic classification, evaluated by grouped cross-validation ( $R^2$ , adjusted  $R^2$ , RMSE, MAE, permutation importance, VIF, AUC/PR-AUC). Supporting analyses use a bootstrap mixture-LRT with AIC/BIC, a GEV extreme-value fit, a Wald sequential probability ratio test (with decision boundaries  $A = \ln \frac{1-\beta}{\alpha}$ ,  $B = \ln \frac{\beta}{1-\alpha}$ , a stopping time, and an expected-sample-size approximation), one- and two-sample nonparametric tests (Wilcoxon signed-rank; Mann-Whitney), an  $F$ -test/Kruskal-Wallis omnibus with post-hoc comparisons, and a  $\chi^2$  independence test with Cramér's  $V$  and standardized residuals. The one-sided test of  $b > 2/3$  for men ( $z = 46.8$ ) is our Neyman-Pearson example: uniformly most powerful in the one-parameter exponential-family idealization, with the GLRT as its composite-hypothesis generalization. The extreme-value fit uses the GEV distribution

$$G(x) = \exp \left\{ - \left( 1 + \xi \frac{x - \mu}{\sigma} \right)^{-1/\xi} \right\}, \quad (2)$$

where  $\xi < 0$  implies a bounded upper tail (a strength ceiling) and  $\xi \geq 0$  does not; we estimate  $\xi$  from equal-size block maxima and bootstrap its interval. The sequential test's

The two-component shape of raw Total tracks sex; Dots normalization removes it

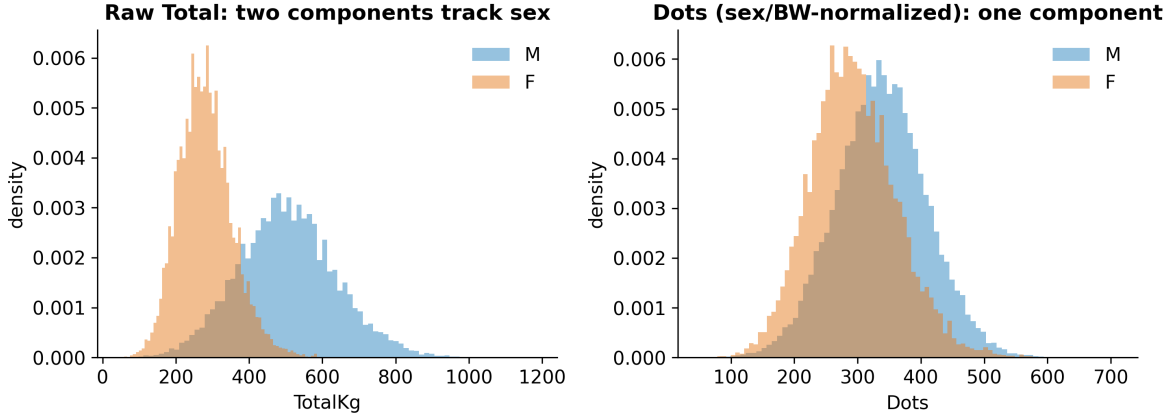


Fig. 6. The apparent “mixture” in raw Total is sex; Dots removes it.

Sequential test (stopping time) and a-priori power

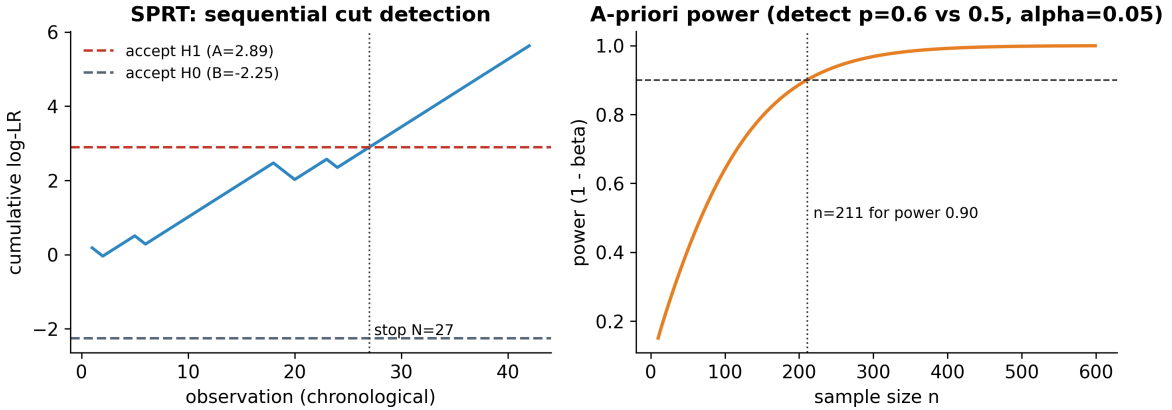


Fig. 7. Sequential test (stopping time  $N = 27$ ) and a-priori power ( $n = 211$  for power 0.90).

expected sample size under the alternative is approximated by  $\mathbb{E}[N | H_1] \approx [(1 - \beta)A + \beta B] / [p_1 s_1 + (1 - p_1)s_0]$ , with  $s_0, s_1$  the per-observation log-likelihood increments. Table II maps the course toolbox onto the study.

**Reproducibility.** Every result is produced by a small set of scripts from the pinned snapshot; numbers are written to machine-readable files, figures are exported at 300 DPI, and an independent re-derivation from the raw data reproduced every headline value reported here. All analyses use Python 3.12 with numpy/scipy/statsmodels/scikit-learn and the rddensity implementation of the CJM test; exact versions are pinned in the repository.

#### IV. DISCUSSION

**Conclusions.** The two numbers a competitor controls leave measurable, falsification-surviving traces. Attempt loads are quantized to the plate grid; bodyweight bunches specifically below class limits, and not at controls or placebos in the bunching direction. Beyond strategy, the data also bear on the limits of strength: scaling with bodyweight is sex-specific and

departs from the isometric  $2/3$ , and basic meet-record variables already explain  $\sim 70\%$  of the variance in the total. The same dataset supports the descriptive, testing, and prediction analyses.

**Limitations and null results.** This is an observational study: we identify population-level patterns consistent with weight cutting, not the intent of any individual. We report three null results as genuine findings. The 120 kg boundary (the entry to the unbounded 120+ class) does *not* show significant bunching after correction ( $t = -1.6$ ,  $p = 0.11$ ), consistent with weaker cutting pressure at that open-ended boundary; the extreme-value analysis does *not* support a strength ceiling (the  $\xi$  confidence interval includes zero); and we find no material association between tested status and cutting once pseudo-replication is removed. On the methods side, lifter identity is name-based, the per-lifter meet is sampled at random, and the formal bunching test is restricted to the modern kg-only era. The 83 kg signal is robust to the federation set: the de-heaped log-ratio is  $+1.92$  across the IPF-affiliated federations

**H3 residuals: a heavy lower tail (non-normal); the slope is CLT-robust at large  $n$**

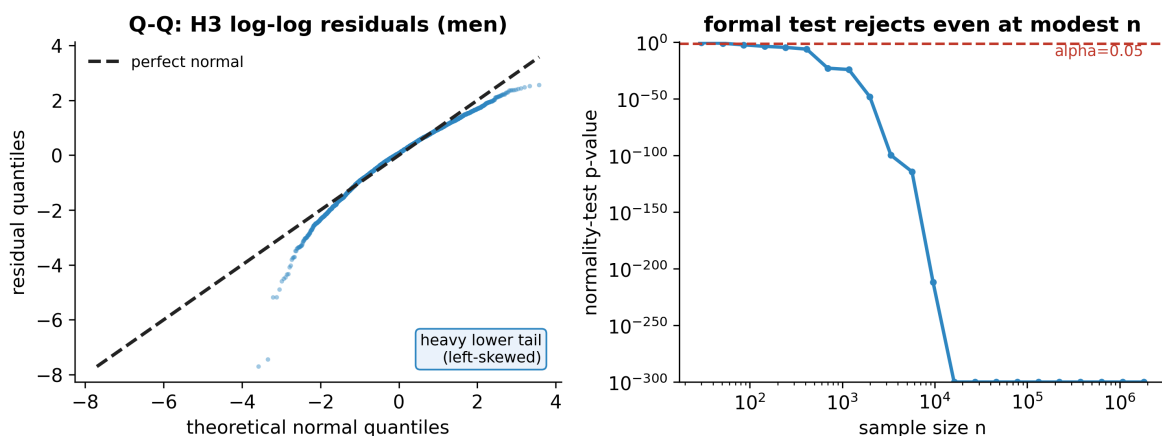


Fig. 8. H3 residuals: heavy lower tail; the slope is CLT-robust at large  $n$ .

TABLE II  
COVERAGE OF THE COURSE TOOLBOX.

| Concept                                | Where used          |
|----------------------------------------|---------------------|
| GLRT, MLE, $\chi^2$ goodness-of-fit    | H1 grid model       |
| Boundary LRT, parametric bootstrap     | H1 null calibration |
| Density discontinuity (McCrary/CJM)    | H2 bunching         |
| Multiple testing (Bonf./Holm/Šidák/BH) | H2 limits, H3       |
| OLS, $R^2$ , Wald, interaction term    | H3 allometry        |
| One-/two-sided, one-sample tests       | H1, H3              |
| Neyman–Pearson MP/UMP framing          | H3 one-sided test   |
| Pearson/Spearman, Fisher CI            | H3                  |
| Regression, random forest, logistic    | H4 prediction       |
| Learning metrics, grouped CV, VIF      | H4                  |
| Mixture LRT, AIC/BIC                   | structure           |
| Welch $t$ / Mann–Whitney (two-sample)  | tested vs. untested |
| Wilcoxon signed-rank (paired)          | attempt 1 vs. 3     |
| $F$ /ANOVA, Kruskal–Wallis, post-hoc   | equipment           |
| $\chi^2$ independence, Cramér’s $V$    | tested $\times$ cut |
| Extreme-value theory (GEV)             | strength ceiling    |
| Sequential Wald (SPRT), stopping time  | cut detection       |
| Power, Type I/II tradeoff              | a-priori power      |

used here and +1.66 when restricted to IPF/USAPL alone. Several extensions remain for future work: more federations and eras, a structural bunching model that recovers the share of manipulators, and a multiverse robustness check.

## REFERENCES

- [1] H. J. Kleven, “Bunching,” *Annual Review of Economics*, vol. 8, pp. 435–464, 2016.
- [2] J. McCrary, “Manipulation of the running variable in the regression discontinuity design: A density test,” *J. Econometrics*, vol. 142, no. 2, pp. 698–714, 2008.
- [3] M. D. Cattaneo, M. Jansson, and X. Ma, “Simple local polynomial density estimators,” *J. Amer. Statist. Assoc.*, vol. 115, no. 531, pp. 1449–1455, 2020.
- [4] E. Franchini, C. J. Brito, and G. G. Artioli, “Weight loss in combat sports: physiological, psychological and performance effects,” *J. Int. Soc. Sports Nutrition*, vol. 9, art. 52, 2012.
- [5] J. H. J. Einmahl and J. R. Magnus, “Records in athletics through extreme-value theory,” *J. Amer. Statist. Assoc.*, vol. 103, no. 484, pp. 1382–1391, 2008.
- [6] J. C. Peyen, R. Nuermaimaiti, and J. Cunningham, “Discussing diminishing returns: A new scoring system for powerlifting,” arXiv:2503.13040, 2025.
- [7] H. Chernoff, “On the distribution of the likelihood ratio,” *Ann. Math. Statist.*, vol. 25, no. 3, pp. 573–578, 1954.
- [8] S. G. Self and K.-Y. Liang, “Asymptotic properties of maximum likelihood estimators and likelihood ratio tests under nonstandard conditions,” *J. Amer. Statist. Assoc.*, vol. 82, no. 398, pp. 605–610, 1987.
- [9] OpenPowerlifting, <https://www.openpowerlifting.org> (data released into the public domain).